

Abivan Vijay

Robotics • Embedded Hardware • Web Applications

Chennai, India · +91 98404 08614 · abivan100@gmail.com · github.com/abivan100-stack · abivan-dev.vercel.app

PROFILE

Student builder who designs and builds ESP32 hardware, autonomous robots and full-stack web applications, and takes them to national competitions, hackathons and expos. Uses AI coding tools (Claude Code, Codex) to take ideas from prototype to working build quickly.

ACHIEVEMENTS

UPCOMING	National Robotics Championship 2026, Delhi: drone soccer and innovation contest	20–21 Nov 2026
UPCOMING	Robotics for Good Youth Challenge (ITU): national round, Delhi	10 Oct 2026
	Robotics for Good Youth Challenge: selected for the national round, Senior category	Sep 2026
	PEC Hacks 4.0: Consolation Prize, Vault (team lead)	Aug 2026
	Young Environmental Scientist 2026 (IWMA with IIT Madras): participant, Sewer Gas Detector	Aug 2026
	School Shark Tank-style pitch competition: presented Volt, with all 14 branches competing	Aug 2026
	National Robotics Championship: selected with CRASH, after qualifying at the SRM zonals	Jul 2026
	School-level expo: Winner, FreshSense	Feb 2026
	RoboWunder International Competition: participant at the Regionals, FreshSense	Jan 2026
	SRM University, Ramapuram expo: participant, Smart Guard	Nov 2025

PROJECTS

CRASH: Chennai Road Accident Safety Hub · team project	JavaScript · Python
Web app that maps road accidents across Greater Chennai, ranks junctions by a severity-weighted risk score and recommends interventions, with analytics dashboards, Monte Carlo simulation and an AI data assistant.	
Robotics for Good: Public-Health Response Robot · team project	LEGO SPIKE Prime
Fully autonomous robot that runs 2-minute simulated emergency-response missions: moving samples, closing a quarantine zone, delivering medical kits and colour-sorting patients.	
Vault: Cold-Chain Integrity Console · team lead	TypeScript · ESP32
Monitors a vaccine shipment's cold chain and records every event to a hash-chained ledger that can be verified afterwards.	
Sewer Gas Detector · team project	ESP32 · ESP-NOW
Two-unit wearable for sanitation workers: reads H ₂ S and methane every second, alarms at set thresholds, and relays to a supervisor unit over ESP-NOW with no internet needed.	
FreshSense: Food Spoilage Detection · team lead, 3 members	ESP32 · Blynk IoT
ESP32 conveyor system that classifies food as FRESH or ROTTEN using gas sensors and a self-calibrated baseline, with remote monitoring through Blynk IoT.	
Volt: Peer-to-Peer Solar Energy Ledger · team project	TypeScript · Python
Transparent, tamper-evident ledger for rooftop-solar energy trading, built for the Open Energy Challenge 2026.	
Also built: Smart Guard (team lead, 3 members), WriteWise AI (Flask + Claude), Landmine Detection Shoe (prototype), RFID Door Lock.	

TECHNICAL SKILLS

Hardware	Arduino, ESP32, LEGO SPIKE Prime; gas, climate and RFID sensors, displays, servos; Wi-Fi, ESP-NOW, Blynk IoT
Software	JavaScript, TypeScript, HTML, Python, Node.js; FastAPI, Flask, Express; MongoDB, Supabase; Claude API
Tools	Claude Code, Codex, Git/GitHub, Render, Vercel

EDUCATION

Velammal Academy, Nolambur, Chennai: Grade 9, CBSE (age 14)	2026–27
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